



Sleep

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Introduction

- Sleep is a basic human need to all individuals. Humans **spend about one-third of their lives asleep.**
- Sleep disorders and sleep deprivation are contributing factors to the development of many chronic diseases, such as type 2 diabetes, heart disease, depression, and obesity.

Definition of sleep

- Sleep is a complex biological rhythm. the individual is awake when the body temperature is highest, and asleep when the body temperature is lowest.
- Circadian regularity begins to develop by the 6th week of life, and by 3 to 6 months most infants have a regular sleep-wake cycle.

We require sleep for many reasons

- **To prevent fatigue**
- **To conserve energy**
- **To restore the mind and body**
- **To enjoy life more fully.**

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- **To enhances daytime functioning**
 - **To vital for cognitive, physiologic, and psychosocial function.**
 - **allows an individual to awaken feeling alert and refreshed**
 - **an important factor in an individual's quality of life**
 - **To cope with daily stresses**

Physiology of Sleep

- Sleep has come to be considered an altered state of consciousness in which the individual's perception of and reaction to the environment are decreased.
- The cyclic nature of sleep is thought to be **controlled by centers located in the lower part of the brain.**

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- **During sleep, the growth hormone is secreted, and cortisol is inhibited. With the beginning of daylight, melatonin is at its lowest level in the body and the stimulating hormone, cortisol, is at its highest.**



N.B.

Some environmental stimuli for sleep Will usually awaken a sleeper, Whereas many other noises will not.

Types of Sleep

- 1.NREM (non-rapid eye-movement) sleep**
- 2.REM (rapid-eye-movement) sleep.**

NREM Sleep

- About **75%** of sleep during a night is NREM sleep.
- It is divided into **three stages**. Each of the stages is associated with distinct brain activity and physiology.

Stages of NREM sleep

Stage 1

- The stage of **very light sleep** and lasts only a few minutes.
- During this stage, the individual **feels drowsy and relaxed**, the **eyes roll from side to side**, and the **heart and respiratory rates drop slightly**.
- The **sleeper** can be readily **awakened** and may **deny that he or she was sleeping**.



Stage 2

- **The eyes are generally still** body processes continue to **slow down; the heart and respiratory rates decrease slightly, and body temperature falls.**
- **An individual in stage 2 requires more intense stimuli than in stage 1 to awaken, such as touching or shaking.**

Stage 3

- The deepest stage of sleep.
- During deep sleep, the sleeper's heart and respiratory rates drop 20% to 30% below those exhibited during waking hours.
- The sleeper is difficult to arouse.
- The skeletal muscles are very relaxed, and snoring is most likely to occur.
- This stage is essential for restoring energy and releasing important growth hormones

Physiologic Changes During NREM Sleep

- **blood pressure falls.**
- **Pulse rate decreases.**
- **Peripheral blood vessels dilate.**
- **Cardiac output decreases.**

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- **Skeletal muscles relax.**
 - **Basal metabolic rate decreases 10% to 30%.**
 - **Growth hormone levels peak.**
 - **Intracranial pressure decreases.**

Clinical Alert!

- ❑ **Sleep deprivation** in hospitalized clients **contributes to delirium.**
- ❑ **Delirium initially presents as agitation, or confusion**

REM sleep

- **Most dreams take place during REM sleep but usually will not be remembered unless the individual arouses briefly at the end of the REM period.**
- **During REM sleep, the brain is highly active, and brain metabolism may increase as much as 20%.**
- **REM sleep usually recurs about every 90 minutes and lasts 5 to 30 minutes.**

Clinical Alert!

- ❑ Clients who experience **sleep deprivation** will more commonly **experience an increase in fatigue.**
- ❑ Sleep deprivation **results in a significant increase in thyroid stimulating hormone, which increases tension, anger, and hostility**

Sleep Cycles

- During a sleep cycle, individuals typically pass through NREM and REM sleep,
- The complete cycle usually lasting about 90 to 110 minutes in adults.
- In the first sleep cycle, a sleeper usually passes through the first two stages of NREM sleep in a total of about 20 to 30 minutes. Stage 3 lasts about 50 to 60 minutes.

Functions of Sleep

1. **Restore normal levels of activity.**
2. **Necessary for protein synthesis, which allows repair processes to occur.**
3. **The role of sleep in psychologic well-being: Individuals with inadequate amounts of sleep tend to become emotionally irritable, have poor concentration, and experience difficulty making decisions.**

Normal Sleep Patterns in different ages

Newborns

- **Newborns sleep 12 to 18 hours a day, on an irregular schedule with periods of 1 to 3 hours spent awake.**

Unlike older children and adults

Infants

- **At first, infants awaken every 3 or 4 hours, eat, and then go back to sleep.**
- **Periods of wakefulness gradually increase during the first months.**
- **By 6 months, most infants sleep through the night (from midnight to 5 a.m.) and begin to establish a pattern of daytime naps.**

Toddlers

- **Between 12 and 14 hours of sleep are recommended for children 1 to 3 years of age.**
- **Most still need an afternoon nap, but the need for midmorning naps gradually decreases.**

Preschoolers

- The preschool-age child (3 to 5 years of age) requires 11 to 13 hours of sleep per night, particularly if the child is in preschool.
- Many children of this age dislike bedtime

School-Age Children

- The school-age child (5 to 12 years of age) needs 10 to 11 hours of sleep per night, but most receive less because of increasing demands (e.g., homework, sports, social activities). They may also be spending more time at the computer and watching TV.



Adolescents

- **Adolescents (12 to 18 years of age) require 8 to 10 hours of sleep each night.**
- **Teens are sleepy at times and in places where they should be fully awake—at school, at home, and on the road. This can result in lower grades, negative moods (e.g., unhappy, sad, tense), and increased potential for car crashes.**

Older Adults

- **Older adults (65 to 75 years) usually awaken 1-3 hours earlier and go to bed approximately 1 hour earlier than younger adults (ages 20 to 30).**
- **They may awaken an average of six times during the night. Although the ability to sleep becomes more difficult.**



Most healthy adults

- **Get 7 to 9 hours of sleep per night.**
- **However, individual needs do vary—some adults may be able to function well (e.g., without sleepiness or drowsiness) with 6 hours of sleep, and others may need 10 hours to function optimally.**

Signs that may indicate that an individual is not getting enough sleep

- **Falling asleep or becoming drowsy during a task**
- **Not being able to concentrate or remember information**
- **Being unreasonably irritable with others.**
- **contributes to short-term memory loss**
- **Inadequate performance on newly learned tasks.**

Factors that contribute to diminished sleep

- **Stress**
- **Depression**
- **Pain**
- **Travel**
- **Lifestyle roles, such as job, student, or parenting.**
- **Adults working long hours.**

Factors Affecting Sleep

- Both the quality and the quantity of sleep are affected by quality factors.
- **Sleep quality:** is a subjective characteristic and is often determined by whether an individual wakes up feeling energetic or not.
- **Quantity of sleep:** is the total time the individual sleeps.

Factors Affecting Sleep

Illness

- Illness that causes pain or physical distress (e.g., back pain) can result in sleep problems.
- Individuals who are ill require more sleep than normal.

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- **The need to urinate during the night disrupts sleep, and individuals who awaken at night to urinate sometimes have difficulty getting back to sleep**

Environment

- **Environment can promote or hinder sleep.**
- **The individual must be able to achieve a state of relaxation prior to entering a period of sleep. Any change, such as noise in the environment, can inhibit sleep.**

Lifestyle

- **Following an irregular morning and nighttime schedule can affect sleep.**
- **Moderate exercise in the morning or early afternoon is usually conducive to sleep, but exercise late in the day can delay sleep.**

Emotional Stress

- Stress is considered by most sleep experts to be the one of the greatest causes of difficulties in falling asleep or staying asleep.
- An individual who becomes preoccupied with personal problems (e.g., school- or family or marriage problems) may be unable to relax sufficiently to get to sleep.

Stimulants and Alcohol

- **Drinking beverages containing caffeine in the afternoon or evening may interfere with sleep.**
- **Individuals who drink an excessive amount of alcohol often find their sleep disturbed.**

Diet

- **Weight gain** has been associated with reduced total sleep time and earlier awakening.
- **Weight loss**, on the other hand, seems to be associated with an increase in total sleep time and less broken sleep.

Smoking

- Nicotine has a stimulating effect on the body, and smokers often have more difficulty falling asleep than nonsmokers.
- Smokers are usually easily aroused and often describe themselves as light sleepers.

Medications

- Some medications **affect the quality of sleep.**
- **Beta blockers** have been known to **cause insomnia and nightmares.**

These drugs may disrupt REM sleep, delay onset of sleep, or decrease sleep time

- **Antidepressants**
- **Bronchodilators**
- **Caffeine**
- **Narcotics**

Common Sleep Disorders

Insomnia

- **Insomnia is described as the inability to fall asleep or remain asleep.**

Types of insomnia

Acute insomnia: Lasts one to several nights and is often caused by personal stressors or worry.

Chronic insomnia

- If the insomnia persists for longer than a month

Chronic-intermittent insomnia Which means difficulty sleeping for a few nights, followed by a few nights of adequate sleep before the problem returns .

Risk factors insomnia

- ❖ **Older age** : The incidence of insomnia **increases with age**,
- ❖ **Medical condition.**
- ❖ **Female gender: women suffer sleep loss in connection with hormonal changes** (e.g., menstruation, pregnancy, and menopause).

Symptoms of Insomnia



- **Difficulty falling asleep**
- **Waking up too early or frequently during the night**
- **Difficulty returning to sleep**
- **Difficulty concentrating**
- **Irritability**
- **Fatigue**
- **Difficulty at work or school**

Treatment for insomnia

- ❑ Develop new behavior patterns that induce sleep and maintain it.
- *Stimulus control*: creating a sleep environment that promotes sleep
- *Cognitive therapy*: learning to develop positive thoughts and beliefs about sleep
- *Sleep restriction*: limiting time in bed in order to get to sleep and stay asleep throughout the night

Medication

- Antihistamines are thought to be safer for older clients, their side effects (i.e., dizziness, and hypotension).
- It should not be recommended for any client with a history of increased intraocular pressure, cardiovascular disease, or hypertension.